



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science and Engineering

"A place to learn; A chance to grow"

2025-26 (Even Semester)

MSE-2 EXAMINATION



Course: B.Tech. In Computer Science and Engineering

Year/Semester: 6th Semester (3rd Year)

Subject: Robotic Automation System

Subject Code: CS6E004D

Max Marks: 20

Date: 04/04/2026

Duration: 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

- CO1: Understand the fundamental principles of robotics and industrial automation.
CO2: Select and apply appropriate sensors, actuators, and control techniques.
CO3: Analyse AVR RISC Microcontroller architecture.
CO4: Develop and simulate ARM Processor.
CO5: Evaluate real-world robotic automation systems and current trends.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Explain servo motor.	L2/CO2	1
Q.2	Enlist the types of memory in AVR.	L2/CO3	1
Q.3	Express an interrupt.	L1/CO4	1
Q.4	Define localization in robotics.	L2/CO1	1

SECTION B

1. All Questions are compulsory.
2. Answer in one two three sentences.

Q.5	Describe proximity sensor.	L2/CO2	2
Q.6	Differentiate between RISC and CISC architectures with reference to AVR microcontrollers.	L4/CO3	2
Q.7	Explain USART communication in AVR.	L2/CO4	2
Q.8	Discuss mapping in robotics.	L1/CO5	2

SECTION C

1. Solve any TWO questions.

Q.9	Describe the architecture of the AVR family microcontrollers with a neat block diagram. Explain the role of CPU, ALU, and control unit.	L2/CO3	4
Q.10	Explain the classification of robots based on configuration and application with examples.	L4/CO2	4
Q.11	Discuss robotic perception and elaborate the role of sensors and AI techniques in enabling robots to interpret their environment.	L2/CO5	4